

AI for Global 4 Notes:

Benefits of AI:

- Faster response/solution times
 - Can auto-classify, route and acknowledge tickets, cutting down human interaction, and could therefore be used to auto assign tickets to a to-do list
 - Self-service/bot-deflection (knowledge base suggestions, FAQ bots) reduce load and could speed up the resolution of simple/trivial tickets
- Improved efficiency and productivity
 - By automating the repetitive work, SecOps can focus on the complex and high-value tickets.
 - Studies and tools show that automating portions of ticket flows can yield returns - BoldDesk says AI help desks can deliver a ROI of 100%-300% dependent on volume and how many interactions can be automated
- Better prioritisation and quality
 - AI can better assess the urgency of a ticket via multiple inputs - the severity, number of affected users, customer importance, SLA (service-level agreements, and similarity to past incidents. This allows for a more accurate ranking of tickets
 - Sentiment analysis and text analysis could help detect frustrations or crises that need escalating
- Scalability and handling peak loads
 - During load spikes (e.g. total software outages, high season, campaigns), the AI can help to reduce backlog explosion.
 - Since the AI tool can work 24/7, it could enable support functions to operate outside of business hours or in other time zones more cheaply.
- Consistency, reduced errors and better analytics.
 - AI reduces classification or routing errors (human mistakes, mis-tags).
 - Having structured metadata and model feedback can provide insights into ticket trends, root causes, and common failure types. You can then use that to improve product reliability, show the fault in a product and therefore reduce incoming tickets
- Improved customer satisfaction.
 - Faster replies, fewer back-and-forths, consistency in who they're communicating with, being routed to the correct specialist all will contribute to a better service.
- Better SecOps experience and retention
 - Removing tedious tasks helps to avoid burnout and lets the team do what they do best.
 - Tools that assist with summarisation and suggested responses can reduce cognitive load.

Case Studies:

- Everise
 - Using Retell AI, they 'contained' (resolved/prevented escalation of) 65% of internal service desk tickets with the bot, translating into roughly 600 man-hours saved per month.

- Academic examples
 - TaDaa, a real time ticket assignment bot, achieved ~95.2% accuracy on suggesting where a ticket should be sent, and ~79% accuracy top-5 accuracy for recommending the correct resolver.
 - ESSMaRT (Brightsquid) cut down about 9.2 minutes per user request on average by automating summarisation, escalation prediction, etc.

Risks and Challenges

- Quality and accuracy issues
 - If AI mis-classifies, mis-routes or gives inappropriate suggestions, this could lead to slow resolution, so will require human oversight.
- Context and domain expertise
 - It may struggle with ambiguous/domain-specific tickets, so training data must reflect real tickets, SOPs, customer types.
- User trust and transparency
 - Customers and SecOps may not trust the automated decisions - it should explain why and also have easy overrides.
- Data security
 - Tickets may have sensitive data. It must comply with GDPR, HIPAA etc, mask PII, secure storage
- Edge cases
 - Highly urgent or complex issues, emotional or escalated customers, etc., might need human interaction or special pathways.

Flow:

- Incoming ticket/alert
- AI organiser
 - Reads the text and metadata
 - Extracts issue type, urgency, customer importance
 - Scores priority (0-100)
 - Suggests category and assignee
- Database Layer
 - Stores ticket and AI metadata
 - Tracks updates (status, SLA, agent feedback)
- UI/UX Layer
 - Displays priority badges (Low/Med/High/Crit)
 - Shows "Suggested Actions"
 - Highlights upcoming SLA risks.

